SANADA: A Reproducible Synthetic Acoustic Dataset for Intractable Mixtures, Evaluated with Baseline Models in Separation, Counting, and Tagging

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Abstract

Practical acoustic environments contain an unknown, time-varying number of simultaneous sources, yet most separation corpora cap evaluations at $K \leq 4$ speakers. Collecting high-density recordings with isolated ground-truth stems is physically intractable, making a controlled synthetic approach necessary. We present SANADA (Samslesha-Nada): three domain-isolated, reproducible mixture corpora designed for variable-K evaluation. Org-Mix-4 and Org-Mix-9 scale organisational speech from $K = 2$ to $K = 9$; FSD-Mix spans 32 environmental sound classes up to $K = 32$. All corpora enforce pre-mix source partitioning to prevent data leakage, apply -3.0 dBFS logarithmic attenuation to avoid clipping, and maintain full temporal overlap to maximise spectral masking. Baseline experiments cover source counting (Conformer Regression), multi-label tagging (Conformer + Asymmetric Loss) and

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monaural separation (Conv-TasNet). Performance degrades beyond $K = 4$, a threshold we term the *Complexity Ceiling*, with extreme-density separation ultimately proving computationally intractable for standard architectures. SANADA is publicly available via IEEE DataPort (DOI: 10.21227/z7hv-e014).

Keywords: audio source separation, source counting, sound event detection, synthetic dataset, variable-source mixtures, multi-talker speech, deep learning, SANADA

1. Introduction

Speech separation, sound counting, and acoustic tagging are increasingly required to operate together in dense scenes: meeting rooms, lecture halls, and city streets where speakers, transient events, and ambient processes overlap without pause. Solving all three tasks jointly demands that a system simultaneously *separate* constituent sources, *count* active entities, and *recognise* their semantic classes from a single channel. Deep learning has advanced each task individually over the past decade [1]. Permutation invariant training (PIT) [2, 3] and deep clustering [4] established the foundations for talker-independent separation. Conv-TasNet [5] and dual-path RNNs [6] then moved processing entirely into the time domain, yielding large gains on two- and three-speaker benchmarks. More recently, HDemucs [7], the Audio Spectrogram Transformer [8], and Conformer networks [9] extended these gains to classification and regression settings. Each of these architectures, however, has been validated almost exclusively on corpora with a fixed, low speaker count, leaving their behaviour under high source density untested.

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1.1. Structural Gaps in Existing Corpora

Studying separation, counting, and tagging in realistic overlapping scenes requires datasets that span a range of source densities. Core speech separation benchmarks such as WSJ0-2mix/3mix [10], LibriMix [11], WHAM! [12], and WHAMR! [13] provide high-quality mixture–stem pairs and have driven much of the progress in multi-speaker models. Their mixtures, however, are limited to a small, fixed number of speakers, typically $K \leq 3$, so they do not reveal how architectures behave as the number of simultaneous sources increases. Universal and noise-robust corpora such as FUSS [14] extend beyond read speech and support domain-agnostic separation [15, 16], but they still lack a controlled, variable-K progression comparable to conversational settings. Large-scale audio datasets including FSD50K [17], AudioSet [18], and ESC [19] emphasise label diversity and have been effective for sound event tagging, yet they consist mainly of mixed recordings without isolated stems or precise control over source overlap. Without pre-mix reference signals, standard separation metrics such as scale-invariant signal-to-distortion ratio (SI-SDR) [20] cannot be computed reliably in high-density conditions.

1.2. The SANADA Suite & Key Contributions

To address this dataset gap, we introduce **SANADA** (Samslesha-Nada) [21], a reproducible set of synthetic acoustic corpora designed for variable-K mixtures. Org-Mix-4 and Org-Mix-9 cover organisational speech with up to nine simultaneous speakers, while FSD-Mix extends to 32 environmental sound classes. Together they expose regimes in which fixed-output models struggle when the number of active sources is unknown or large. SANADA is built around four design goals:

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1. **Density scaling and domain isolation:** The suite consists of three domain-specific corpora. Org-Mix-4 ($K \in \{2, 3, 4\}$) and Org-Mix-9 ($K \in \{2, \ldots, 9\}$) focus on speech, whereas FSD-Mix ($K \in \{2, 3, 15, 20, 25, 32\}$) is derived from environmental audio. Mixtures are formed only within each domain, so performance changes can be attributed to source density rather than cross-domain mismatch. Each corpus includes fixed-K subsets and variable-K splits, allowing the emergence of a "complexity ceiling" to be measured explicitly.
2. **Temporal overlap and energy control:** For $K \geq 2$, all sources are active over the full 8.0-second window, preserving natural pauses within each stem but avoiding artificial gaps between sources. A constant -3.0 dB attenuation is applied to interfering stems before mixing, preventing digital clipping while preserving relative levels.
3. **Modern baselines across tasks:** We provide reference results with representative models: a Conformer-based regressor for source counting [9], a Conformer with Asymmetric Loss (ASL) for multi-label tagging [22] and Conv-TasNet for monaural separation [5]. These baselines define a clear starting point for future work.
4. **Monaural stress test:** All mixtures are single-channel. This forces models to rely purely on spectral and temporal cues, without spatial information, and makes SANADA a conservative stress test before moving to multi-microphone extensions.

1.3. Problem Setting

We focus on single-channel, multi-source mixtures with an unknown and time-varying number of active sources (variable-K). Each 8.0-second wave-

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form may contain speech or environmental sounds from several classes. Depending on the task, the objective is to (i) estimate K (source counting), (ii) predict which classes are present (multi-label tagging), or (iii) reconstruct an isolated waveform for each source (separation). All SANADA experiments are defined within this setting. Furthermore, because SANADA releases explicit, time-aligned unmixed master stems prior to polyphonic superposition, the corpus extends beyond multi-source regression and separation, naturally supporting individual source classification, acoustic feature profiling, and monaural speech recognition evaluations.

2. Related Work

Conv-TasNet [5], SepFormer [24], and MossFormer2 [25] represent a clear architectural progression: from time-frequency masking toward time-domain convolution and transformer-based attention. Yet the benchmarks used to evaluate these models have not kept pace. Most corpora fix K at two or three sources, so the behaviour of these architectures under higher source counts remains largely unknown.

2.1. Deep Learning for Speech and Audio Source Separation

Speech separation corpora such as LibriMix [11] and WHAMR! [13] mainly provide mixtures with two or three speakers. They are therefore not representative of organisational or environmental scenes in which the number of active sources K is unknown and can be much larger. At the other extreme, large-scale tagging datasets such as AudioSet [18] offer broad class coverage but do not include isolated stems, making them unsuitable for evaluating separation metrics. Many existing benchmarks also emphasise

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clip-level classification over instance-level reconstruction, so they provide limited insight into how models behave when several sources of the same class overlap.

2.2. Speech Mixture Datasets

Most work on monaural speech separation has been driven by a small set of low-density corpora. Two-speaker (WSJ0-2mix) and three-speaker (WSJ0-3mix) mixtures, derived from WSJ0 [26], are the standard testbeds for PIT, deep clustering, and Conv-TasNet-style models [2, 4, 5]. LibriMix [11] extends this setup with additional noise conditions and sampling rates, while WHAM! [12] and WHAMR! [13] add background noise and room reverberation. In all of these corpora, however, the number of simultaneous speakers is fixed at $K = 2$ or $K = 3$, so they do not support systematic evaluation of separation models as the speaker count increases to higher values such as $K = 8$ or $K = 9$.

2.3. General Audio Tagging and Sound Event Datasets

Large-scale audio datasets are typically designed for classification rather than controlled mixture modelling. FSD50K [17] and FSDKaggle2018 [27], both derived from Freesound [23], provide single-channel clips with dense labels across many classes using the AudioSet ontology [18]. These resources have been central to recent progress in sound event tagging, but they do not supply isolated stems or explicit control over the number of overlapping sources. Environmental benchmarks such as ESC-50 [19] and domestic scene datasets such as DESED [28] share the same limitation. As a result, they

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are not suitable for joint evaluation of separation, counting, and tagging as a function of K.

2.4. Universal and Music Separation Datasets

Universal separation datasets attempt to cross acoustic domains within a unified generation framework. FUSS [14] synthesizes mixtures of arbitrary sound events with varying degrees of overlap, serving as the primary benchmark for domain-agnostic separation networks [15, 16]. However, FUSS is heavily geared toward generic sound extraction and lacks the controlled, high-density variable-K progression required to simulate crowded conversational or organizational speech scenarios. In the music domain, MUSDB18 [29] provides multitrack mixtures and stems, but is strictly constrained to a small, fixed taxonomy of instruments (vocals, drums, bass, accompaniment). Slakh [30] extends this concept using synthetic MIDI-derived audio, enabling robust training data ablations. While invaluable for algorithmic music separation, these corpora do not encompass speech or broad environmental domains, nor do they scale K dynamically to simulate chaotic, unconstrained acoustic interference. Crucially, while existing standard corpora like MUSDB18 [29] and FUSS [14] arbitrarily plateau at maximum densities of $K \leq 4$, SANADA dynamically scales to $K = 32$, explicitly isolating the extreme-density variable-K gap. A comprehensive comparison of these architectural limits is detailed in Table 1, the proposed Org-Mix-4/Org-Mix-9 and FSD-Mix suite resolves these critical deficits. By providing variable-K mixtures scaling up to $K = 32$, paired with absolute ground-truth stems and unified metadata across both speech and sound event domains, SANADA establishes a rigorous continuum for high-density machine listening.

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Table 1: Comparison of representative existing datasets with the proposed suite. "Stems" indicates isolated source availability; "Var.-K" indicates an explicit architectural design for variable and unknown active source counts.

Dataset	Domain	K Range	Stems	Var.-K
WSJ0-2mix/3mix [26, 4]	Speech	2, 3	Yes	No
LibriMix [11]	Speech	2, 3	Yes	No
WHAM(R)! [12, 13]	Noisy speech	2	Yes	No
FSD50K [17]	Gen. audio	–	No	–
FSDKaggle18 [27]	Gen. audio	–	No	–
AudioSet [18]	Gen. audio	–	No	–
MUSDB18 [29]	Music	≤ 4	Yes	No
FUSS [14]	Gen. audio	Variable	Yes	Partly
Org-Mix-4 (Ours)	Org. speech	2–4	**Yes**	**Yes**
Org-Mix-9 (Ours)	Org. speech	2–9	**Yes**	**Yes**
FSD-Mix (Ours)	32-class audio	2–32	**Yes**	**Yes**

3. Dataset Organization and Curation

The SANADA suite is hosted on IEEE DataPort and organised for direct use with common deep learning frameworks such as PyTorch [31]. In addition to the final mixtures, the repository includes the raw recordings and the processed pre-mix stems. All audio is provided as single-channel 16 kHz, 16-bit PCM WAV files with a fixed duration of 8.0 seconds.

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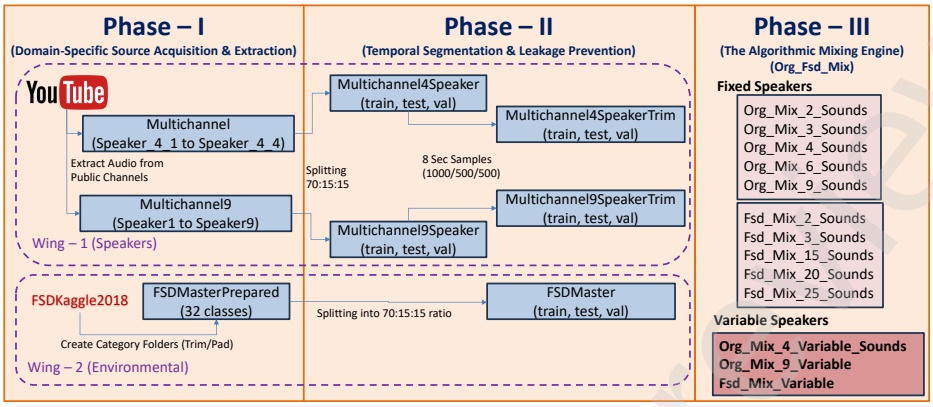


Figure 1: SANADA generative pipeline. Phase 1 partitions raw recordings into domain-specific source pools. Phase 2 applies pre-mix segmentation and data splits. Phase 3 performs domain-isolated mixing with fixed attenuation.

3.1. Repository Architecture and Data Flow

The repository is organised into three phases that mirror the generative pipeline. As shown in Fig. 1, raw audio is first grouped into three independent master pools (FSD, 4-speaker, and 9-speaker) at the isolated-source level. This separation is preserved through the later mixing stages, so speech and environmental audio are never combined across domains.

3.2. Directory Breakdown (User Guide)

3.2.1. Raw Input Directories

The `Multichannel`, `Multichannel9`, and `FSDKaggle2018` directories contain the original, variable-length recordings from public organisational lectures and the Freesound database [23, 27]. These files are included for provenance and reproducibility.

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3.2.2. Isolated Ground-Truth Stems (`*Trim` and `FSD_Master`)

These folders hold the stem pools used by the mixing engine. The 70/15/15 train/validation/test split is applied at this level to avoid data leakage between sets. The pools are kept separate by domain:

- **Speech stems (4-speaker pool):** `Multichannel4SpeakersTrim` contains 1,000 training, 500 validation, and 500 test clips per speaker, each 8.0 seconds long.
- **Speech stems (9-speaker pool):** `Multichannel9SpeakersTrim` follows the same quotas for the nine speaker identities.
- **FSD stems:** `FSD_Master` groups 8.0-second clips into 32 environmental classes (e.g., *Acoustic_ guitar, Glockenspiel, Gunshot_ or_ gunfire, Laughter*).

3.2.3. Final Generated Mixtures (`Org_Fsd_Mix`)

`Org_Fsd_Mix` stores the 14 mixture datasets derived from these stem pools. Mixtures are created only within a domain: speech stems are not mixed with FSD stems, and the 4-speaker and 9-speaker pools are not combined. Within each domain, the repository provides fixed-K subdirectories (every clip has exactly K sources, e.g., `FSD_Mix_2_Sounds`) and `*Variable` subdirectories (where K varies across clips). Each mixture corpus contains 1,000 training, 100 validation, and 100 test mixtures.

3.3. Unified Metadata Manifests

Each train, validation, and test subdirectory in `Org_Fsd_Mix` includes a `metadata.csv` file. This manifest links every mixture waveform to its labels,

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source stems, and basic energy statistics through the following fields:

- `mixture_path`: absolute path to the generated 8.0-second mixture.
- `speaker_labels` / `fsd_labels`: list of active class labels for the mixture.
- `num_sources`: number of overlapping sources (K).
- `source_files`: paths to the isolated stems used to create the mixture.
- `mixture_dBFS`, `mixture_peak_dBFS`: overall level and peak level of the mixture.

3.4. Dataset Footprint and Statistics

Across its three domain-specific corpora, the SANADA suite contains 27,112 pre-mix stems (60.25 hours, 6.47 GB) and 15,600 synthesized mixtures (34.66 hours, 3.72 GB). When the raw recordings are included, the full repository occupies 28.27 GB. Table 2 lists stem and mixture counts for each corpus. To assess the acoustic quality of the organisational speech stems, Waveform Amplitude Distribution Analysis (WADA-SNR) [32] was applied to all Org-Mix clips. The pool yielded a mean SNR of 18.4 ± 2.1 dB and an RT60 of $\approx$0.47 s, indicating that the 15-second boundary buffer removed most opening and closing artefacts while preserving near-anechoic vocal content.

3.5. Class Imbalance and the Case for Synthetic Mixing

Open-domain audio collections show a pronounced long-tail class distribution [17]. As Fig. 2 illustrates, sustained sounds (*Acoustic_guitar*, *Laughter*)

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Table 2: Per-corpus stem pool and synthesized mixture statistics for the SANADA suite. K-range lists the fixed-density values plus a variable split (Var).

Corpus	K**-Range**	**Stems (pre-mix)**		**Mixtures (synthesized)**		
		Files	**Duration**	**Files**	**Duration**	**Size**
Org-Mix-4	$\{2, 3, 4, \text{Var}\}$	8,000	17.78 h	∼4,800	10.66 h	1.14 GB
Org-Mix-9	$\{6, 9, \text{Var}\}$	18,000	40.00 h	∼3,600	8.00 h	0.86 GB
FSD-Mix	$\{2, 3, 15, 20, 25, \text{Var}\}$	1,112	2.47 h	∼7,200	16.00 h	1.72 GB
Total	**13 sub-datasets**	**27,112**	**60.25 h**	**∼15,600**	**34.66 h**	**3.72 GB**

contribute many more valid 8-second stems to the `FSD_Master` pool than transient events (*Shatter*, *Keys_jangling*). Evaluating models directly on such unbalanced, crowdsourced data mixes two factors: acoustic density and class frequency. The SANADA mixer draws stems uniformly over classes, so performance differences mainly reflect source-separation difficulty rather than label frequency. This makes the "Complexity Ceiling" easier to study as a function of K.

4. Generative Pipeline & Acoustic Mixing Methodology

The SANADA pipeline generates mixtures for both fixed-K and variable-K settings while keeping conditions comparable across speech and environmental domains. Because the source material comes from web lecture audio and crowdsourced environmental recordings, the design emphasises avoiding data leakage and unintended cross-domain mixing. As outlined in Fig. 1, the process is organised into three stages: domain-specific source acquisition (Phase 1), pre-mix partitioning and temporal windowing (Phase 2), and single-channel mixing with controlled overlap and attenuation (Phase 3).

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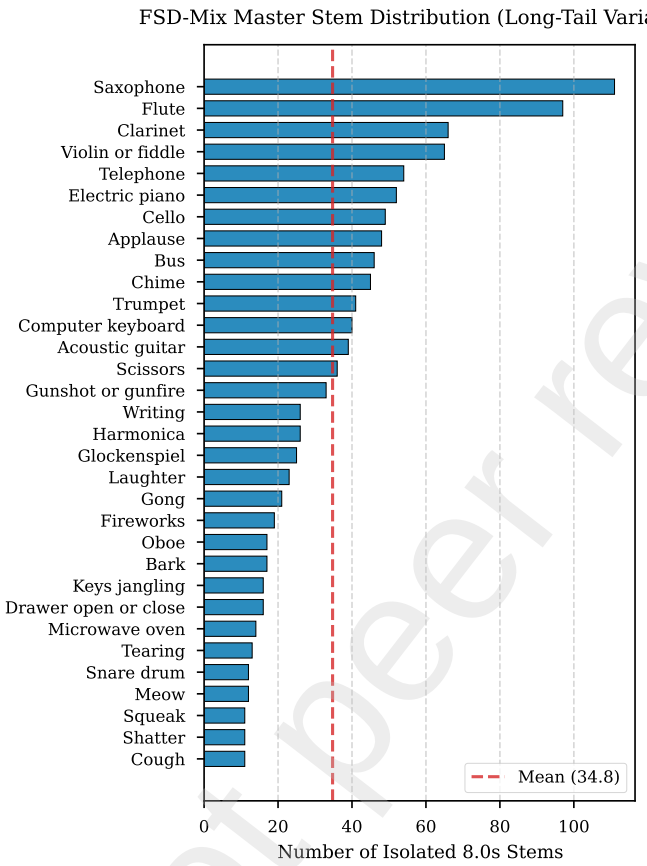


Figure 2: Stem counts per class in `FSD_Master` (32 classes). Sustained sounds dominate; transient events are scarce.

4.1. Phase 1: Domain-Specific Source Acquisition & Extraction

4.1.1. Wing 1: Organisational Speech Corpus

The speech stems for the organisational mixtures are taken from publicly available lecture videos on YouTube, following a common practice in large-scale speech datasets such as VoxCeleb [33]. Only recordings with close-talking, studio-style microphones are retained to limit reverberation and background noise. Two speaker pools are created:

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- **4-speaker pool:** stored in `Multichannel/`, with subdirectories `Speaker_4_1` to `Speaker_4_4`.
- **9-speaker pool:** stored in `Multichannel9/`, with subdirectories `Speaker1` to `Speaker9`.

Ethical considerations and privacy: All speakers gave consent for research use of their voice data. Identifying metadata is removed, and speakers are referred to only by anonymous labels (for example, `Speaker_1` to `Speaker_9`). **Automated extraction and normalisation:** To avoid manual variation, Wing 1 extraction is run through a Python script. Audio is downloaded with `yt-dlp`, using authenticated access to respect rate limits, and decoded with `FFmpeg` to handle container and codec differences. The `librosa` library [34] converts each file to a single-channel, 16 kHz, 16-bit PCM `.wav` file, as summarised in Algorithm 1.

4.1.2. Wing 2: Open-Domain Environmental Sounds

The second wing ingests environmental sounds from `FSDKaggle2018.audio_train`, using the accompanying `FSDKaggle2018.meta` annotations as supervision [27, 17]. Compared with the lecture recordings, these clips vary widely in quality, length, and label reliability, so an explicit curation step is applied. Curated clips are written into 32 class directories under `FSD_Master_Prepared/`. The classes are *Acoustic_guitar, Applause, Bark, Bus, Cello, Chime, Clarinet, Computer_keyboard, Cough, Drawer_open_or_close, Electric_piano, Fireworks, Flute, Glockenspiel, Gong, Gunshot_or_gunfire, Harmonica, Keys_jangling, Laughter, Meow, Microwave_oven, Oboe, Saxophone, Scissors, Shatter, Snare_drum, Squeak, Tearing, Telephone, Trumpet, Violin_or_fiddle, Writing.* The cura-

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Algorithm 1 Wing 1: Organisational Speech Extraction and Standardisation

Require: Target playlist URL, target $SR = 16000$

Ensure: 16 kHz, single-channel WAV files

Step 1: Download audio
Initialise `yt-dlp` with session cookies and rate limiting
for each *video* **in** playlist **do**
    Extract best audio stream → `raw_audio.mp3`
end for
Step 2: Transcode and standardise
for each $file$ **in** extracted directory **do**
    Repair container and decode with `FFmpeg`
    $y, sr \leftarrow$ librosa.load($file$, sr $= SR$, mono $=$ True)
    $out_path \leftarrow$ replace_extension($file$, ".wav")
    soundfile.write(out_path, y, SR, PCM_16)
end for

tion script enforces:

- **Label reliability:** only clips with `manually_verified = 1` are used.
- **Duration constraints:** clips shorter than 6.0 s are discarded; clips longer than 8.0 s are truncated; clips between 6.0 and 8.0 s are zero-padded with silence to exactly 8.0 s (128,000 samples at 16 kHz).
- **Format standardisation:** all retained clips are resampled to 16 kHz, converted to single-channel, and saved as 16-bit PCM `.wav` files.
- **Class coverage:** any class with fewer than 10 valid clips after filtering

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is removed.

Algorithm 2 outlines this process.

4.2. Phase 2: Temporal Segmentation & Leakage Prevention

After source acquisition, SANADA applies dataset splitting and temporal windowing before any mixing takes place. Many synthetic benchmarks create mixtures first and then divide them into train, validation, and test sets; this post-mix splitting allows the same speaker or event instance to influence multiple splits and can inflate reported metrics [35, 11]. In SANADA, partitioning is performed at the raw-file level before trimming or mixing.

4.2.1. Wing 1: Pre-Mix Data Partitioning

For organisational speech, the long-form audio in `Multichannel/` and `Multichannel9/` is divided using a fixed random seed. To support reproducible experiments across frameworks [31], the pseudo-random number generator is initialised with $S = 42$. For each speaker directory (for example, `Speaker_4_1` or `Speaker1`), the script shuffles the list of files and applies a 70/15/15 split for train/validation/test. Files are then copied into `Multichannel4Speakers/` and `Multichannel9Speakers/`, each with separate `train/`, `val/`, and `test/` subdirectories. Any mixture generated for the test set therefore uses utterances that never appear in the training or validation sets. Algorithm 3 summarises this procedure.

4.2.2. Wing 1: Temporal Windowing and Artifact Exclusion

Once the file-level split is fixed, the lecture recordings (typically around 15 minutes long) are converted into fixed-length segments for model training [31]. Boundary regions of web audio often contain music, applause, or

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silence, which can introduce artefacts. To reduce this, a 15.0-second buffer is applied at the start and end of each file. Valid 8.0-second segments are drawn by sampling a random start time within this central region. Each segment is resampled to 16 kHz, converted to mono, and saved as a 16-bit PCM WAV file, following common practice in separation work [1, 5]. The process repeats until 1,000 training, 500 validation, and 500 test clips are obtained for each speaker. The resulting stems are stored in `Multichannel4SpeakersTrim/` and `Multichannel9SpeakersTrim/`, preserving the train/val/test structure from the previous step. Algorithm 4 describes this procedure.

4.2.3. Wing 2: Environmental Sound Partitioning

The FSD-based clips were already trimmed or padded to 8.0 s in Phase 1, so no additional windowing is needed. The `FSD_Master_Prepared/` directory therefore acts as the source for a file-level split in the same way as the speech data [23]. The partitioning script reuses the protocol from Algorithm 3 with the same seed ($S = 42$), iterating over the 32 class folders and assigning files to train/validation/test in a 70/15/15 ratio. The resulting stems are written to `FSD_Master/` with separate `train/`, `val/`, and `test/` subdirectories. Any variable-K FSD mixture for the test set is created only from the 15% of clips reserved for testing, so evaluation on sound event tagging and universal separation is carried out on unseen examples [15, 14].

4.3. Phase 3: Algorithmic Mixing Engine

The final stage of the pipeline generates polyphonic waveforms from the partitioned stem pools. Simple linear mixing without constraints can lead to unrealistic overlap patterns and frequent clipping, so the SANADA mixer

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uses explicit rules for overlap, cardinality, and level control [11, 14].

4.3.1. Algorithmic Superposition and Overlap

In dense environments such as meetings or streets, sources often overlap for long periods and mask each other spectrally [13]. For all mixtures with $K \geq 2$, SANADA keeps every source active across the full 8.0-second window. To construct a mixture with cardinality K, the engine first samples K distinct classes or speakers from the relevant pool and then selects one 8.0-second stem per class. The stems are time-aligned from start to finish, but the instantaneous overlap still varies because each stem retains its original pauses, onsets, and decays. This produces mixtures with realistic sparsity patterns while ensuring a consistent global overlap condition.

4.3.2. Acoustic Attenuation and Clipping Prevention

As K increases (up to 9 speakers in speech and 32 classes in FSD-Mix), summing stems at full scale can cause integer overflow in 16-bit PCM. Instead of dynamic range compression, which would change the spectral envelope and affect metrics such as SI-SDR, SANADA applies a fixed attenuation to interfering sources. Let $\mathcal{S} = \{s_1(t), s_2(t), \ldots, s_K(t)\}$ be the selected stems. The first stem $s_1(t)$ is used as the base. Each additional stem is scaled by -3.0 dB before being added:

$$x_{\text{mix}}(t) = s_1(t) + \sum_{i=2}^{K} 10^{\frac{-3.0}{20}} \, s_i(t), \quad t \in [0, T]$$

This fixed gain reduces the risk of clipping while keeping the relative levels between sources approximately intact.

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4.3.3. Unified Metadata During Mixing

During mixing, the engine also writes a `metadata.csv` file (Algorithm 5). For each mixture it records the output path, the list of active labels, the value of K, the source stem paths, and summary level statistics (average dBFS and peak dBFS), providing a single ground-truth entry usable across separation, counting, and tagging tasks [31, 36].

4.3.4. Mixture Quotas and Combinatorial Scale

The stem pools created in Phase 2 are large (for example, 1,000/500/500 stems per speaker), and enumerating all combinations would produce an impractically large corpus. Instead, the mixer draws a fixed number of mixtures per configuration. For each dataset variant (e.g., `Org-Mix-4`, `Org-Mix-9`, `FSD-Mix-Variable`), the engine generates:

- 1,000 training mixtures (from the 70% training stems),
- 100 validation mixtures (from the 15% validation stems),
- 100 test mixtures (from the 15% test stems).

This scheme keeps train/validation/test completely disjoint, while providing consistent dataset sizes across all tasks.

4.4. Acoustic Characteristics and Density Analysis

To characterise the acoustic conditions created by SANADA, we computed macro-level DSP statistics over both the isolated stems and the variable-K mixtures. Following [37], we report mean spectral centroid (a proxy for spectral "brightness") and root mean square (RMS) energy for each partition in Table 3. The transition from single-source stems to dense mixtures

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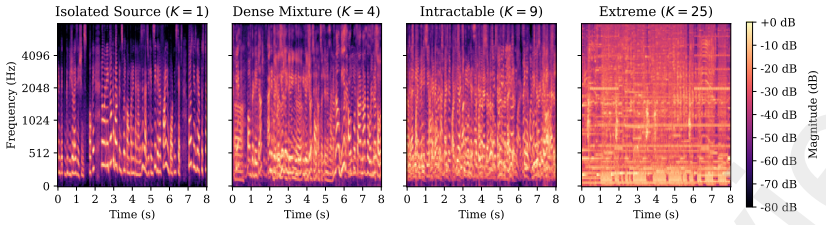


Figure 3: Mel-spectrogram examples at increasing source counts: an isolated stem ($K = 1$), a four-source mixture ($K = 4$), a high-density speech mixture ($K = 9$), and an environmental mixture with $K = 25$. Higher K values reduce visible temporal gaps and increase spectral overlap.

produces the expected changes in these measures. For example, between `Multichannel9` stems and the highest-density speech mixture (`Org_Mix_9_Sounds_Variable`), the mean spectral centroid decreases from 2112.8 Hz to 1503.5 Hz, consistent with strong spectral masking when up to nine speakers overlap. Over the same transition, RMS energy increases with K, reflecting the accumulation of multiple sources. Figure 3 illustrates this trend spectrally, showing how temporal sparsity visible at $K = 1$ is largely lost in mixtures with $K = 9$ and $K = 25$.

5. Baseline Experiments & Results

We report baseline performance on SANADA for source counting, multi-label tagging and universal source separation [15, 14]. Across all tasks and domains, model accuracy declines consistently as the number of overlapping sources K increases, showing that the SANADA mixtures cover a regime that is challenging for current architectures. To ensure comparability across all three tasks and datasets, all models were trained for a fixed number of epochs: 30 epochs for source counting and multi-label tagging, and 10

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Table 3: Acoustic characteristics of SANADA single-source stems and variable-K mixtures.

Dataset Partition	Mean RMS Energy	Mean Spectral Centroid (Hz)
Isolated Stems (Single Source)		
Multichannel	0.0787 ± 0.0413	2957.0 ± 538.0
Multichannel9	0.0662 ± 0.0618	2112.8 ± 470.0
FSDKaggle2018.audio_train	0.0586 ± 0.0717	4113.7 ± 2503.8
FSD_Master	0.0568 ± 0.0606	1861.8 ± 970.9
Synthesized Mixtures (Variable K)		
Org_Mix_4_Sounds_Variable	0.1509 ± 0.0459	1694.9 ± 141.4
Org_Mix_9_Sounds_Variable	0.2040 ± 0.0804	1503.5 ± 133.3
FSD_Mix_Variable_Sounds	0.2162 ± 0.0946	2147.1 ± 443.1

epochs for monaural separation, chosen to maintain training stability across the variable-K density regimes.

5.1. Multi-Source Density Estimation (Counting)

Estimating the number of active sources K is an important step for separation systems that must handle variable source counts in real recordings [15, 14]. As a baseline, we use a Conformer-based regressor [9]: the classification head is replaced by a single regression output trained with mean squared error (MSE). The network predicts a real-valued count that is rounded to the nearest integer at inference time. Results are summarised in Table 4 and Fig. 4. On Org-Mix-4, the model can track source counts reliably, achieving 74.75% exact accuracy and 100% relaxed accuracy (within ± 1). Moving to Org-Mix-9 reduces exact accuracy to 36.46%, and on the 32-class FSD-Mix setting the MAE rises to 6.73 with only 5.56% exact and 14.44% relaxed accuracy. These results suggest that counting becomes substantially harder as both the maximum K and the class diversity increase.

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Table 4: Source counting baselines with Conformer regression.

Dataset	K_{max}	MAE	RMSE	Exact Acc.	Relaxed Acc (± 1)
Org-Mix-4	4	0.3243	0.4302	74.75%	100.00%
Org-Mix-9	9	0.8766	1.0925	36.46%	84.38%
FSD-Mix-32	32	6.7262	8.0885	5.56%	14.44%

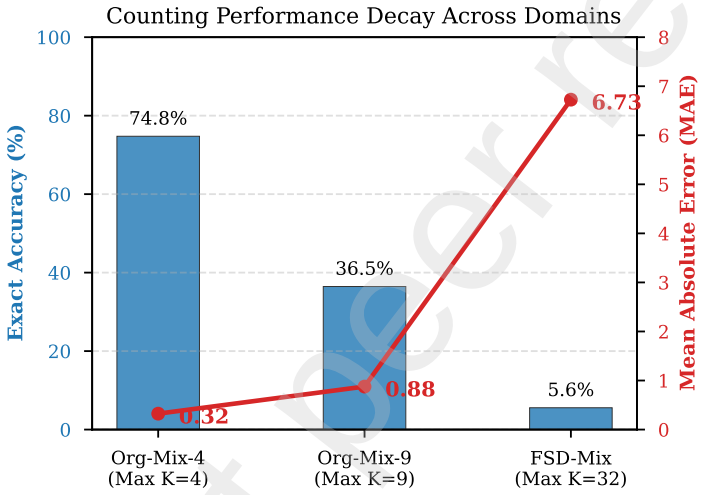


Figure 4: Conformer regression performance on variable-K source counting. As the maximum source count in the dataset increases from 4 to 32, exact accuracy drops and mean absolute error (MAE) grows.

At high densities, SANADA mixtures have little temporal sparsity, so most time–frequency regions contain energy from several sources at once [1]. In this regime, even a strong sequence model such as the Conformer struggles to estimate K from monaural mixtures, reinforcing the need for architectures that can reason explicitly about variable and large source counts.

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5.2. Multi-Label Acoustic Tagging

In addition to separating and counting sources, it is useful to assess how well models can identify which classes are present in a mixture. We evaluate a Conformer-based tagger [9] trained with Asymmetric Loss (ASL) [22] for multi-label acoustic presence detection [27]. Table 5 reports macro-averaged mean average precision (mAP), F1, and ROC-AUC on the three SANADA tagging benchmarks. ASL is designed to handle the imbalance between positive and negative labels in multi-label settings [22], and works well for the lower-complexity tasks: on Org-Mix-4, the model reaches 0.99 macro mAP and 0.94 macro F1. As the number of classes and potential overlaps grows, performance drops. On Org-Mix-9, F1 falls to 0.83; on the 32-class FSD-Mix benchmark, macro mAP decreases to 0.63 and F1 to 0.68. Fig. 5 summarises this trend. With many classes and dense overlap, even detecting which events are present becomes difficult, since most time–frequency regions contain contributions from multiple sources [28].

Table 5: Multi-label tagging baselines with Conformer + ASL.

Dataset	**Classes**	**mAP (Macro)**	**F1 (Macro)**	**ROC-AUC**
Org-Mix-4	4	0.9902	0.9449	0.9765
Org-Mix-9	9	0.9122	0.8305	0.8744
FSD-Mix	32	0.6265	0.6849	0.6161

5.3. Universal Source Separation Performance

The primary objective of source separation is to reconstruct individual isolated signals from a highly overlapped mixture waveform. To establish

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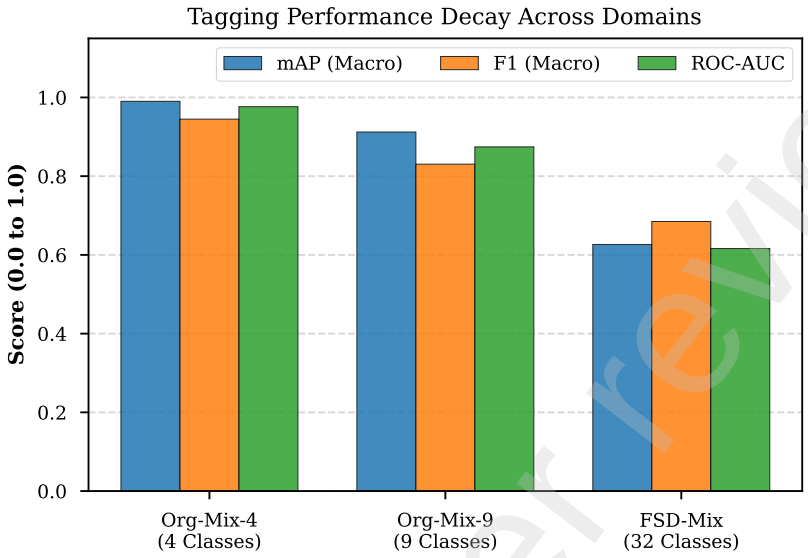


Figure 5: Multi-label tagging performance for Conformer + ASL as the number of classes increases from 4 to 32. All three metrics (mAP, F1, ROC-AUC) decrease with dataset complexity.

a baseline for monaural separation on the SANADA benchmark, we evaluated Conv-TasNet [5], a widely adopted time-domain architecture. Separation fidelity is quantified using the scale-invariant signal-to-distortion ratio (SI-SDR) [20], alongside standard SDR, SIR, and SAR metrics. Our experimental framework evaluates both static and dynamic density conditions, covering fixed source counts of $K \in \{2, 3, 4, 6, 9\}$ as well as variable mixtures containing up to $K \leq 4$ and $K \leq 9$ active sources. The separation networks were trained from scratch using utterance-level permutation invariant training (uPIT) [3] via a `PITLossWrapper` to resolve output-source assignments. Table 6 and Fig. 6 reveal a sharp degradation in separation quality as mixture density increases. Under low-density conditions (Org-Mix, fixed $K = 2$), the model achieves an acceptable SI-SDR of 10.59 dB. However,

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Table 6: Source separation baselines with Conv-TasNet on fixed and variable density mixtures.

Dataset	K Sources	SI-SDR (dB)	SDR (dB)	SIR (dB)	SAR (dB)
Org-Mix (Fixed)	2	10.59	10.98	19.44	11.76
	3	5.08	5.67	12.61	7.14
	4	4.24	4.87	11.74	6.48
	6	-3.40	-2.26	2.95	2.24
	9	-8.57	-6.71	-2.43	1.30
FSD-Mix (Fixed)	2	5.05	6.16	10.20	10.33
	3	0.73	3.14	10.58	8.42
Org-Mix (Variable)	≤ 4	1.98	2.59	7.42	6.06
	≤ 9	-9.61	-7.43	-4.28	2.56
	≥ 10	-	-	-	-

performance collapses at higher densities, with SI-SDR falling below 0 dB beyond $K = 4$ for speech and $K = 3$ for environmental sounds. Crucially, our testing exposed a strict computational and architectural ceiling imposed by the separation paradigm itself. The `PITLossWrapper` successfully executed up to $K = 9$ targets. However, when evaluated on extreme-density mixtures ($K \in \{15, 20, 25\}$), the baseline models fundamentally failed. Because uPIT calculates loss across all possible output permutations, its computational complexity scales factorially as $\mathcal{O}(K!)$. At $K = 15$, computing over 1.3 trillion permutations triggered catastrophic memory exhaustion, failing even on state-of-the-art NVIDIA A100 and H100 GPU environments. Consequently, $K = 9$ represents a hard hardware limit for standard PIT-based training. Furthermore, the introduction of variable-K subsets highlights severe structural vulnerabilities in current architectures. Standard models assume a fixed number of output nodes. To adapt Conv-TasNet for the

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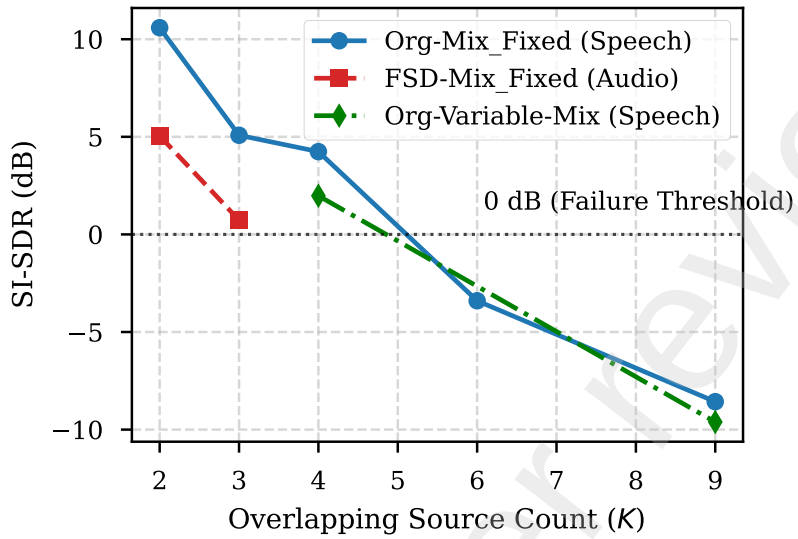


Figure 6: Conv-TasNet SI-SDR on SANADA as a function of K. Performance drops sharply as the number of overlapping sources increases, and further degrades under variable-K conditions.

variable subsets, the model was configured to output the maximum possible sources (e.g., $N_{outputs} = 9$ for Org-Mix-9-Variable), with ground-truth matrices padded with zero-tensors for inactive tracks (e.g., if only 2 sources are active, 7 tracks are zeroed). This creates profound assignment ambiguity: the model struggles to determine *which* 2 of the 9 available output nodes should contain the active signals, leading to substantial performance drops. For instance, while fixed $K = 4$ achieves 4.24 dB, the variable $K \leq 4$ subset plummets to 1.98 dB. Because this is a benchmark dataset paper, our objective is not to modify existing baseline architectures to solve these issues, as that falls outside the current scope. Instead, SANADA serves to explicitly expose these latent vulnerabilities. By demonstrating how zero-tensor padding and factorial loss combinations break current state-of-the-art archi-

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tectures, SANADA provides the necessary empirical foundation for future research into dynamic-cardinality and query-based separation models.

6. Discussion and Limitations

The SANADA suite is intended to extend existing speech and general-audio benchmarks into regimes with high source density and variable K, where many current deep learning architectures are less well tested.[9, 5]

6.1. Strengths and the Complexity Ceiling

A central strength of SANADA is its controlled generative pipeline, which provides explicit source counts, multi-label annotations, and perfectly aligned mixture–stem pairs for every clip. As shown in Section 5, attention-based architectures such as the Conformer maintain good performance at lower densities but exhibit a clear drop in separation, counting, and tagging quality as K increases.[9] For Conv-TasNet in particular, SI-SDR falls sharply beyond $K = 4$ in speech mixtures, even though the model is known to perform well on standard two- and three-speaker benchmarks. [5] This behaviour reflects a structural property of fixed-output architectures: they assume a known, constant number of sources at training and inference time. In contrast, SANADA includes both fixed-K and variable-K splits, so the number of active sources can change from clip to clip. The resulting gap points towards architectures that can handle variable cardinality more directly, such as query-based or target-conditioned separation [39, 38] and models designed for unknown K.[40]

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6.2. Limitations and Future Work

While SANADA provides a useful stress test, several limitations should be noted:

- **Monaural mixtures:** The current release only contains single-channel audio. Many deployed systems rely on microphone arrays and spatial processing for separation and denoising, so a future extension will consider multi-channel versions of the same scenes.[1]
- **Linguistic and acoustic coverage:** The organisational speech wing uses English lecture recordings and does not yet include conversational speech, multiple languages, or a wide range of room impulse responses.[13] Extending SANADA in these directions would broaden its applicability.

Within these constraints, SANADA offers a reproducible, well-documented setting for studying separation, counting, and tagging under high acoustic density and variable source counts.

7. Conclusion

This paper introduced SANADA, a collection of three datasets designed to probe source separation, source counting, and multi-label tagging in high-density acoustic scenes. By combining organisational speech (`Org-Mix-4` and `Org-Mix-9`) with diverse environmental mixtures (`FSD-Mix`), the suite covers source counts from $K = 2$ to $K = 32$ within a single, unified framework. A key component is the generative pipeline, which applies train/validation/test partitioning at the isolated-source level before any mixing, so that evaluation

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uses speakers and events that are unseen during training. To provide baselines, we evaluated Conv-TasNet for monaural separation [5] and Conformer-based models [9] for counting and tagging, using scale-invariant SDR [20] and standard classification metrics. The use of Asymmetric Loss (ASL) for tagging [22] helps in low- and medium-density settings, but all three tasks show a marked loss of performance as K increases, with separation metrics turning negative and counting errors growing with density. Crucially, the separation baselines expose fundamental architectural bottlenecks: standard models struggle under variable-K conditions due to zero-padding ambiguity, while the $\mathcal{O}(K!)$ complexity of permutation invariant training renders extreme-density evaluation ($K \geq 15$) physically intractable. SANADA is made available as an open benchmark to support work on models that can handle large and variable numbers of sources, including target-conditioned and query-based approaches. Future extensions will explore multi-channel versions of the same scenes and training strategies that progress from low to high density within a single framework.

Acknowledgment

The authors thank SR University, Warangal, and NGIT, Hyderabad, for providing institutional support and research infrastructure for this work. The SANADA dataset suite and the accompanying mixing pipeline are publicly available via IEEE DataPort (DOI: 10.21227/z7hv-e014). Generative AI tools were used to assist with language editing and document organisation; all dataset design, experiments, and technical interpretation are the authors' own work, and the authors are responsible for the content of this paper.

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Algorithm 2 Wing 2: Environmental Sound Curation and Standardisation

Require: `FSDKaggle2018.audio_train` directory, `FSDKaggle2018.meta` CSV

Ensure: `FSD_Master_Prepared/` with 32 curated class folders

$df \leftarrow$ Read_CSV(Metadata)
$df \leftarrow df[df.\text{manually_verified} == 1]$
$df \leftarrow$ Filter_Classes_Under_Threshold($df,\ \text{min} = 10$)
for each row **in** df **do**
  $audio \leftarrow$ librosa.load($row.fname,\ \text{sr} = 16000,\ \text{mono} = \text{True}$)
  $duration \leftarrow$ length($audio$)
  if $duration < 6.0$ s **then**
    continue *% discard short clips*
  else if $duration > 8.0$ s **then**
    $audio \leftarrow audio[0 : 8.0\ \text{s}]$
  else
    $pad \leftarrow 8.0\ \text{s} - duration$
    $audio \leftarrow$ Zero_Pad($audio,\ pad$)
  end if
  $class_dir \leftarrow$ Create_Dir("FSD_Master_Prepared/" + $row.label$)
  soundfile.write($class_dir + row.fname,\ audio,\ 16000$)
end for
Post-processing
for each $class_folder$ **in** `FSD_Master_Prepared/` **do**
  if Count_WAVs($class_folder$) < 10 **then**
    Delete_Directory($class_folder$)
  end if
end for

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Algorithm 3 Universal Pre-Mix Data Partitioning

Require: Src_Dir, Out_Dir, ratios: $Train = 0.70$, $Val = 0.15$, $Test = 0.15$

Ensure: `train/`, `val/`, `test/` directories with disjoint files

Initialise deterministic PRNG: random.seed(42)
$Valid_Exts \leftarrow$ {".wav", ".m4a", ".mp3", ".flac"}
for each $Category$ **in** Src_Dir **do**
  $Files \leftarrow$ Get_All_Files($Category, Valid_Exts$)
  if length($Files$) == 0 **then**
    continue
  end if
  random.shuffle($Files$)
  $N \leftarrow$ length($Files$)
  $idx_{train} \leftarrow \lfloor N \times Train \rfloor$
  $idx_{val} \leftarrow idx_{train} + \lfloor N \times Val \rfloor$
  $Subset_{train} \leftarrow Files[0 : idx_{train}]$
  $Subset_{val} \leftarrow Files[idx_{train} : idx_{val}]$
  $Subset_{test} \leftarrow Files[idx_{val} : N]$
  Copy_Preserving_Meta($Subset_{train}, Out_Dir$ + "/train/" + $Category$)
  Copy_Preserving_Meta($Subset_{val}, Out_Dir$ + "/val/" + $Category$)
  Copy_Preserving_Meta($Subset_{test}, Out_Dir$ + "/test/" + $Category$)
end for

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Algorithm 4 Wing 1: Speech Temporal Windowing

Require: Pre-split speaker directory, target quota N, target $SR = 16000$

Ensure: N segments (16 kHz, 8.0 s, mono)

$Buffer \leftarrow 15.0$ % seconds
$Window \leftarrow 8.0$ % seconds
$Samples \leftarrow 0$
while $Samples < N$ **do**
  $file \leftarrow$ Random_Choice(Speaker_Files)
  $T_{total} \leftarrow$ Get_Duration($file$)
  $Limit_{start} \leftarrow Buffer$
  $Limit_{end} \leftarrow T_{total} - Buffer - Window$
  if $Limit_{end} \leq Limit_{start}$ **then**
    continue % file too short
  end if
  $t_{start} \leftarrow$ Uniform_Random($Limit_{start}, Limit_{end}$)
  $y, sr \leftarrow$ librosa.load($file,\ sr = SR,\ $ mono = True,
        offset $= t_{start},\ $ duration $= Window$)
  $out_path \leftarrow$ Format_Name(Speaker, $Samples$) + ".wav"
  soundfile.write($out_path,\ y,\ SR,\ $ PCM_16)
  $Samples \leftarrow Samples + 1$
end while

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Algorithm 5 Phase 3: SANADA Mixing Engine

Require: Split stem directory (train/val/test), target mixtures N, target K

Ensure: Mixture WAV files and `metadata.csv`

Initialise CSV with header fields
$Mixes_Generated \leftarrow 0$
while $Mixes_Generated < N$ **do**
  $\mathcal{C} \leftarrow$ Sample_Distinct_Classes(K)
  $\mathcal{S} \leftarrow [\,]$
  for each c **in** $\mathcal{C}$ **do**
    Append Random_Stem(c) to $\mathcal{S}$
  end for
  $audio_{mix} \leftarrow \mathcal{S}[0]$
  for $i = 1$ **to** $K - 1$ **do**
    $next_stem \leftarrow \mathcal{S}[i]$
    $next_stem \leftarrow$ Apply_Gain($next_stem$, -3.0 dB)
    $audio_{mix} \leftarrow$ Overlay($audio_{mix}, next_stem$)
  end for
  $dest \leftarrow$ Format_Path($Mixes_Generated$)
  Export($audio_{mix}$, $dest$, "wav")
  $dBFS \leftarrow$ Calculate_Average_Energy($audio_{mix}$)
  $Peak_dBFS \leftarrow$ Calculate_Max_Amplitude($audio_{mix}$)
  Append_To_CSV($dest$, $\mathcal{C}$, K, $\mathcal{S}$, $dBFS$, $Peak_dBFS$)
  $Mixes_Generated \leftarrow Mixes_Generated + 1$
end while

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